

Phase locked loop

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Abstract—Characterising the components of a phase locked loop circuit. Finding the phase detector to have the relationship $V_{phase} = (0.4900 \pm 0.0198)/\pi |\Delta\varphi|$ and the VCO: $f_{vco} = (-339.5 \pm 2.1)V_{phase} + (2395.9 \pm 5.9)$. Demonstrating its uses as a frequency multiplier and frequency demodulator.

I. INTRODUCTION

THE phase locked loop (PLL) is a wonderful example of the implementation of negative feedback in circuitry. Synchronisation of electronic oscillators was discovered in 1919 by W.H. Eccles and J.H. Vincent [1]. They observed that, analogous to the synchronisation of weakly coupled pendulums; two electronic oscillators coupled by a resonant circuit would match frequency after a short time.

The first phase locked loops were developed in the early 1930's [2] as an alternative method to demodulate FM signals to the superheterodyne. Initially without phase detectors the oscillator tended to 'drift' in frequency with time. A phase detector was added to add a correction voltage keeping the oscillator at the correct frequency.

However, their use extends beyond that of frequency demodulation: for example in frequency synthesis. They are used commonly in the clock in a CPU to multiple the frequency of a crystal oscillator from Mhz to GHz.

II. THEORY

The PLL circuit consists of two main parts: the phase detector and the voltage controlled oscillator. In order to understand the circuit and their role in its function we consider them separately.

An XOR based phase detector applies the boolean *XOR* operation on two inputs. If the the input signals are both square waves with peak voltage = V_{cc} with a duty cycle of 50% then it can be shown that the output will be a square wave with duty cycle proportional to the phase difference between the inputs.

Feeding this signal through a low pass filter of time constant significantly lower than the input frequency allows us to extract the time averaged integral of the signal. Which we call V_{phase} being therefore proportional to the phase difference between the signals.

$$V_{phase} = k|\Delta\varphi| \quad (1)$$

Where k is the proportionality constant and $\Delta\varphi$ is the phase difference between the inputs.

Using the trivial edge cases $\Delta\varphi = 0$ and $\Delta\varphi = \pi$, we see zero voltage when the inputs are in phase. When they are out of phase we see a constant voltage equal to the peak voltage of the inputs.

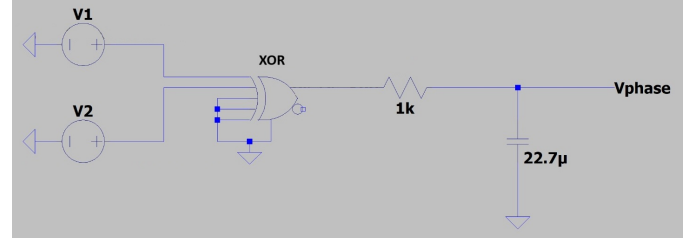


Fig. 1. The LTSpice schematic for the phase detector with low pass filter in the circuit.

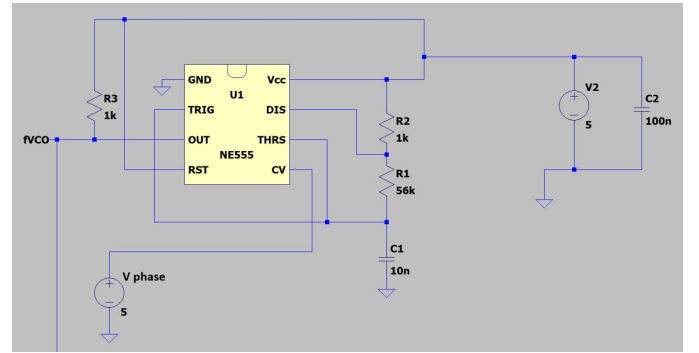


Fig. 2. The LTSpice schematic for the VCO in the circuit with a 555 timer governing its output frequency.

$$V_{phase} = \frac{V_{cc}}{\pi} |\Delta\varphi| \quad (2)$$

Perhaps the most interesting behaviour of this component occurs when its inputs have slightly different frequencies. This causes a rate of change of phase difference causing V_{phase} to sweep across voltages crucial later in establishing a lock.

The Voltage Controlled Oscillator (VCO) is a slightly more complicated component. It outputs a square wave (duty 50%) with a frequency some function of the input voltage called the "control voltage" (CV). The VCO in this circuit uses a 555 timer setup. [3] Where an RC circuit charges and discharges between $CV/2$ and CV . The square wave output comes from this oscillating voltage being compared to a threshold at which point the output becomes V_{cc} or zero.

$$f_{vco} = GV_{phase} + H \quad (3)$$

Where G is a negative constant and H is the base frequency output at 0 input. The value of G and H determined by the time constant of the RC circuit and the second supply voltage labelled $V2$ in 2. The values of G and H will be determined experimentally.

What occurs when the output of the VCO is plugged into the second input of the phase detector is that a short period

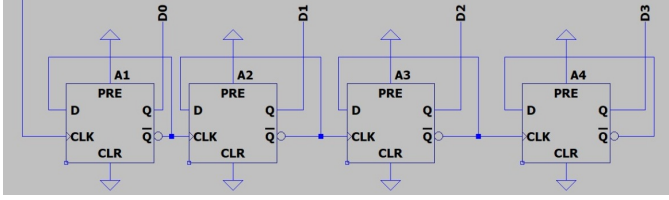


Fig. 3. The LTSpice schematic for the frequency divider in the circuit. Made up of four flip-flops wired in succession acting as a counter.

of time after starting a square wave signal into the first input, V_{phase} sweeps across a range of voltages and settles (with the low pass filter charging and discharging slight each cycle) on a value that causes the VCO output frequency to match that of the first input.

At this point the circuit is "Locked".

A third part of the circuit can be added. 3 Adding a flip flop counter between the VCO's output and the second phase detector input can effectively divide the signals frequency by an arbitrary power of two. E.g. taking the non-inverted output of the fourth flip flop divides f_{vco} by 2^4 (16).

Provided the VCO is sufficiently behaved in the new frequency range the same principles of negative feedback apply and we can still achieve lock. In this case the signal post divider matches the frequency of the input, meaning that f_{vco} must be 16 times greater.

III. METHODS

To build the circuit we used a university supplied breadboard, power supply, and oscilloscope. The components used were the 'TLC 555 CMOS Timer', 'SNx4HC86 Quadruple 2-Input XOR Gates', and two 'SNx4HC74 Dual D-Type Positive-Edge-Triggered Flip-Flops With Clear and Preset'. 4 Data analysis was done in python using *matplotlib*, *numpy* and, *Scipy*.

Whilst we attempted to keep the wires as flat as possible to minimise electrical interference with the signal (see Biot-Savart's law and Faraday's law). There were inevitably some loops of current likely to contribute to noise and uncertainty when taking readings.

Furthermore, the components should have been placed closer together to shorten the wires and minimise the magnetic field induced, neglected in our experiment in part due to our limited experience in circuit building.

Oscilloscope probes were used in place of standard coaxial cables to decrease the load added to the circuit. By including a $9M\Omega$ resistor in its design it attenuates the input signal to the oscilloscope by 10 times. [4] Compensation correction was applied matching the probe's capacitance to the oscilloscope. Stopping it acting as a low-pass filter giving it a mainly flat frequency response therefore higher bandwidth. [5]

The circuit was built with a colour scheme in mind. Red was voltage coming into the circuit, and black was ground. This allowed quick assessments of components and wiring as the signal flow was obvious.

In an attempt to reduce noise in the circuit, three decoupling capacitors were added to the circuit all of capacitance $0.1\mu F$.

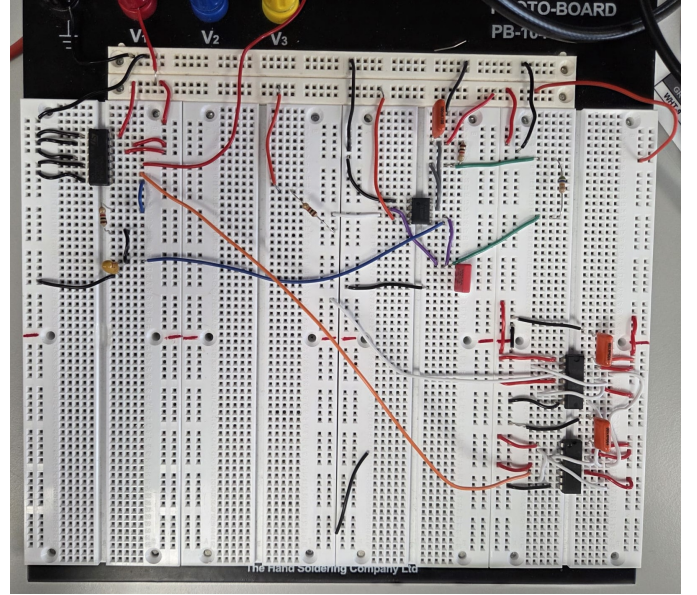


Fig. 4. Top down photo of the circuit as constructed in the lab. The 3 main parts include the phase detector in the top left, connected by the blue wire to the VCO in the top right. Then fed into the counter below, finally coming back to the phase detector through the orange wire.

Two beside the frequency divider and one by the VCO. They were placed as close as possible to the power pins of the components to maximise their effect.

IV. RESULTS & DISCUSSION

To begin with the characterisation of components. The conditions for testing were a complete PLL circuit with 16x divider, recording V_{phase} and the phase difference between V_{vco} and the first input to the phase detector.

A. Phase detector

V_{phase} and $|\Delta\phi|$ were recorded for several input frequencies to the PLL. We expect a straight line graph plotting the two with a zero intercept, obtaining 8 two things were made clear. The first was that a systematic error was present in the data, causing a shift upwards of (0.184 ± 0.011) V. (Uncertainty obtained from the covariance matrix of the linear fit given the present uncertainties in the data.)

The second was that the strength of the fit was strong $R^2 > 0.999$. V_{cc} from the gradient parameter was equal to (0.4900 ± 0.0198) V. Just a 2% difference compared to 2. Whilst this does lie outside the theoretical range, the difference can be attributed to unaccounted load in the circuit as the phase detector was connected across the entire PLL. Perhaps testing the component in isolation would yield better results.

It is worth noting that the visual lack of error bars in 8 is explained by the magnitude of uncertainty being too low to show. 0.01 degrees for the phase difference and 0.001 for V_{phase} .

B. Voltage Controlled Oscillator

Frequency and V_{phase} were taken for another several input frequencies to the PLL. Plotting them gives us: 5. Here, the values of G and H from 3 are found to be.

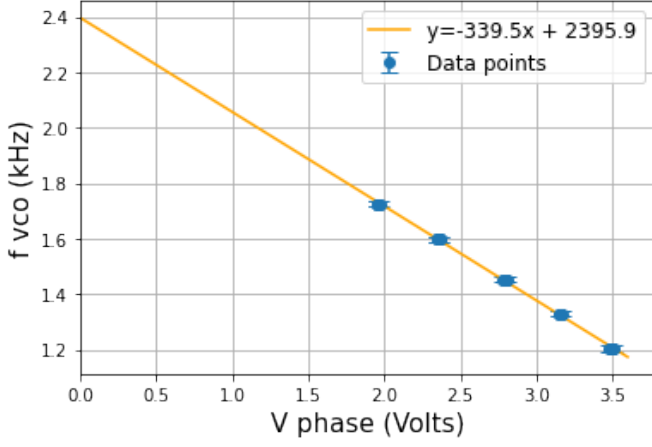


Fig. 5. Plot of f_{vco} against its input: V_{phase} . Clear linear behaviour and strong 1st order polynomial fit.

Symbol	Quantity	Measured value
G	Gradient	-339.5 ± 2.1 Hz/V
H	Base frequency	2395.9 ± 5.9 Hz
G	Gradient (LTSpice)	-313.1 ± 0.0 Hz/V
H	Base frequency (LTSpice)	2367.4 ± 0.0 Hz

C. Testing the whole PLL with frequency synthesis

After connecting up the whole circuit, V_{phase} was measured against time in the moments after the input signal was turned on producing: 6. It demonstrates a successful lock, resembling damped harmonic motion due to the nature of its negative feedback.

After the lock, the two inputs to the phase detector must have a constant phase difference thus equal frequency. Therefore assuming a stable lock, the frequency of the VCO's output must be N times greater than the input frequency given a frequency divider of N between the two.

Below, this behaviour is demonstrated with a 16x divider.

Input Frequency (Hz)	f_{vco} (Hz)	Ratio Output to Input
75	1203.42 ± 0.01	16.04
83	1329.49 ± 0.01	16.01
91	1452.15 ± 0.01	15.95
100	1597.61 ± 0.01	15.97
108	1726.28 ± 0.01	15.98
Symbol	Quantity	Measured value
N	freq multiplication	15.99 ± 0.03

D. Frequency demodulation

The carrier frequency of the square wave was 100 Hz. With a triangular deviation of frequency 1 Hz and deviation of 13 Hz. Figure 7 shows V_{phase} when this FM signal was applied to the PLL. Investigating how accurately it extracts this signal, the fit agreed with the frequency with an error of 64 μ Hz.

The signal (in blue) appears to be highly noisy, however zooming in: shown in the latter figure 7 reveals the typical charging and discharging in the low pass filter. Increasing the time constant of the LPF could smoothen out the noise. However this would cause the PLL to be more sluggish. Increasing the time it takes to lock. And limiting the rate

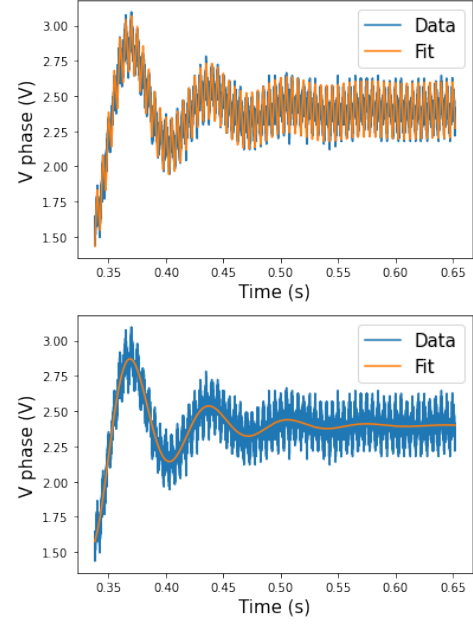


Fig. 6. Plots demonstrating the locking behaviour of the PLL circuit. The figure on top includes a triangular wave term in the fit to account for the low pass filter charging and discharging. Its frequency is twice that of the input frequency, consistent with a phase difference $|\Delta\phi|$ of 90 degrees.

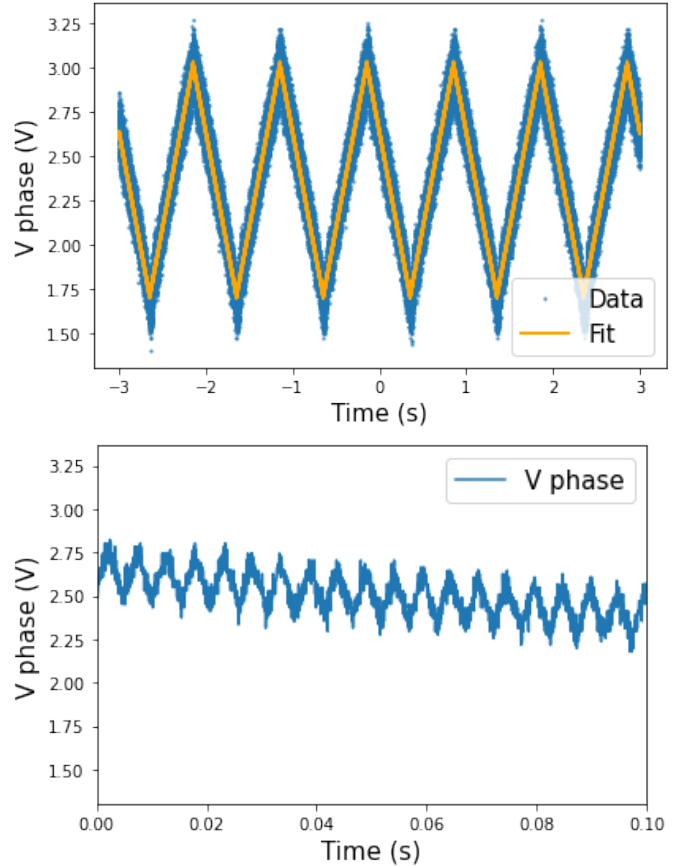


Fig. 7. The upper figure showing V_{phase} when faced with a frequency modulated input signal. It demonstrates its capability at demodulating signals. The lower figure is an expanded view of a short section of time. It shows that a smaller triangular oscillation is ever present in the demodulated signal.

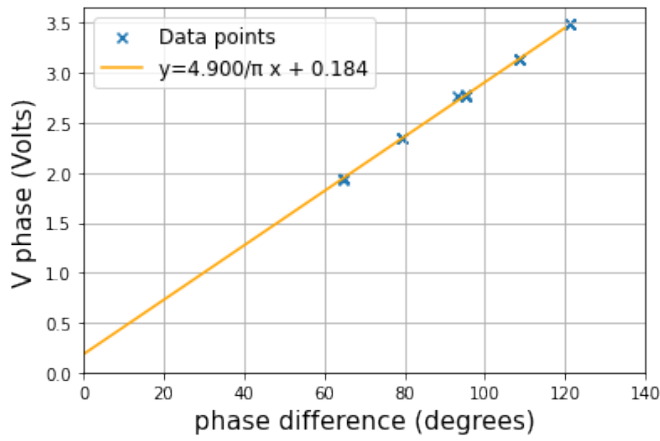


Fig. 8. Plot of V_{phase} against phase difference for the phase detector.

of change of frequency input it can track. Therefore it is probably best to clean up the signal after being received and demodulated, perhaps by passing it through more filters to attenuate the fast oscillating noise.

V. CONCLUSION

Exploring the components of and uses for the PLL circuit. The phase detector and VCO were both found to have linear mappings between their inputs and outputs. The phase detector having a characteristic equation of $V_{phase} = \frac{(0.4900 \pm 0.0198)}{\pi} |\Delta\varphi|$. With a systematic (zero) error of (0.184 ± 0.011) V. The VCO was characterised by $G = (-339.5 \pm 2.1)$ Hz/V and $H = (2395.9 \pm 5.9)$ Hz. Its ability to lock phase, usefulness as a frequency multiplier and use as a frequency demodulator were all demonstrated.

ACKNOWLEDGMENT

ChatGPT was used to generate digital simulations of the PLL and components to better understand them.

REFERENCES

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